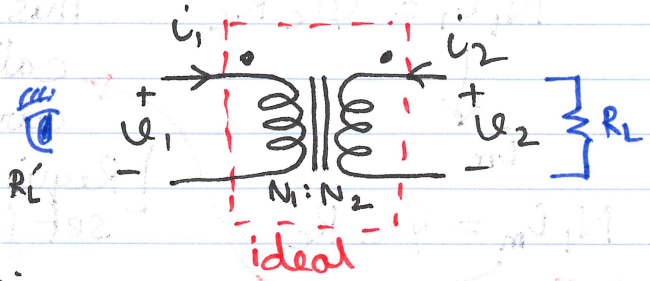
Lecture 14.

Last time: Transformers (Ideal)

Applications:
- Scaling voltage magnitudes.
- impedance matching.



Electrical circuit model of an ideal transformer

Assumptions:
1. infinitely permeable core.
2. No leakage
3. No resistance.

(a) $\frac{v_1}{v_2} = \frac{N_1}{N_2}$

(b) $N_1 i_1 + N_2 i_2 = 0$

(c) $R_L = \frac{v_2}{-i_2} = \left(\frac{N_1}{N_2}\right)^2 \frac{v_1}{i_1}$ (N_2 over N_1)

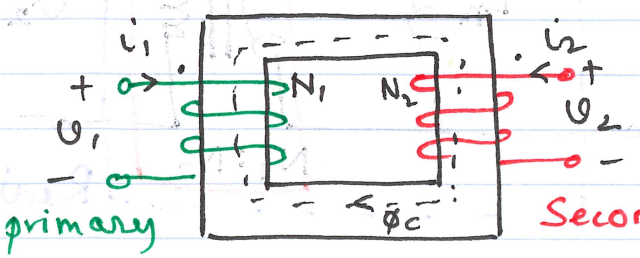
or, $\frac{v_1}{i_1} = \left(\frac{N_1}{N_2}\right)^2 R_L$. ($N_1$ over N_2)

Today: Q1. Electrical circuit model of a real transformer.

Q2. Finding parameters of a real transformer (Important because many times you may be given a transformer that someone else built.)

Q1. Electrical circuit model of a real transformer:

First, let us bring finite permeable core.
[Still no leakage, resistance.]



$N_1 i_1 + N_2 i_2 = \phi_c R_c$

If we set, secondary (wdg #2) as open circuit i.e $i_2 = 0$.

$$N_1 i_1 = \phi_c R_c$$

$$\downarrow$$

$$i_m$$

$$N_1 i_m = \phi_c R_c$$

This i_1 is special called as magnetizing current.
[Required to magnetize, set flux in the core]

Also, $v_1 = \boxed{\frac{N_1^2}{R_c}} \frac{di_1}{dt}$

$\downarrow$

magnetizing inductance, L_m.

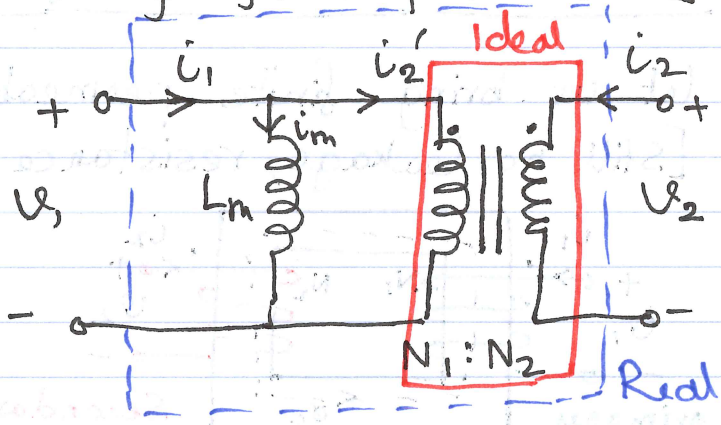
Now, in general, if $i_2 \neq 0$:

$$N_1 i_1 + N_2 i_2 = N_1 i_m$$

$$\therefore \quad i_1 = i_m + \left(-\frac{N_2}{N_1} i_2\right)$$

$+i_2'$ (reflected current on the primary)

$\therefore$ Electrical equivalent circuit of a transformer considering finite permeability of core.



for ideal $N_1 i_2' + N_2 i_2 = 0$

of coupled circuit.

Q. Does our mathematical model still hold true?

$$v_1 = L_m \frac{di_m}{dt}$$

$$= L_m \frac{d}{dt}\left[i_1 - i_2'\right]$$

$$= L_m \frac{di_1}{dt} + \left(L_m \frac{N_2}{N_1}\right)\frac{di_2}{dt}$$

$$= \underbrace{\frac{N_1^2}{R_c}}_{\text{self inductance of wdg}} \frac{di_1}{dt} + \underbrace{\frac{N_1 N_2}{R_c}}_{M\ (\text{mutual inductance})} \frac{di_2}{dt}$$

What about v_2?

voltage scaling across ideal transformer

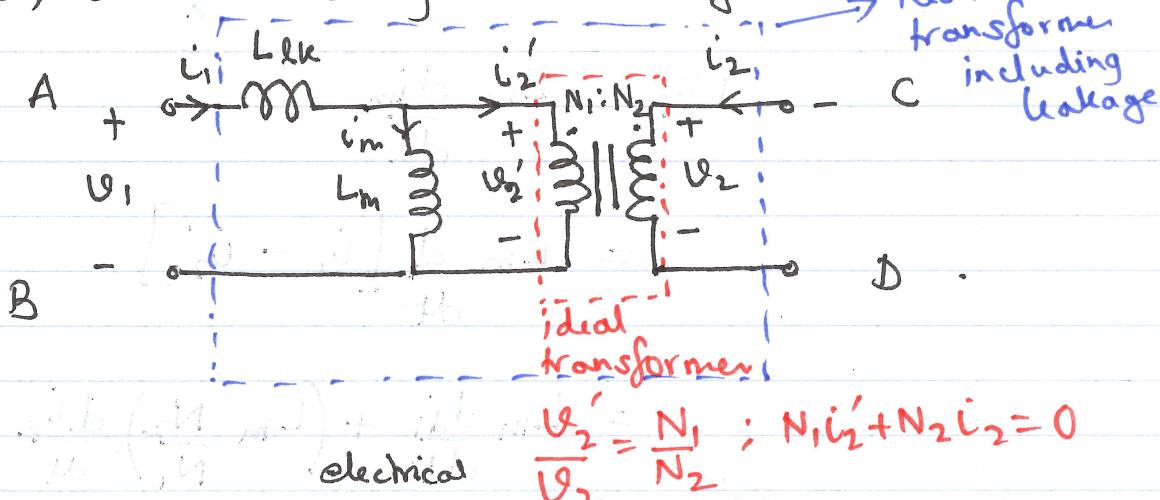
$$v_2 = \frac{N_2}{N_1} L_m \frac{di_m}{dt}$$

$$= \frac{N_2}{N_1} \cdot \frac{N_1^2}{R_c} \cdot \frac{d}{dt}\left(i_1 + \frac{N_2}{N_1} i_2\right)$$

$$= \underbrace{\frac{N_1 N_2}{R_c}}_{M} \frac{di_1}{dt} + \underbrace{\frac{N_2^2}{R_c}}_{\text{self inductance of wdg 2}} \frac{di_2}{dt}$$

Matches our previous mathematical model of our coupled circuit.

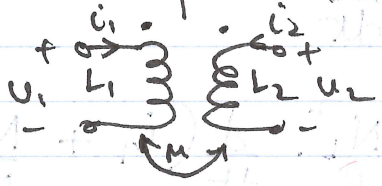
Second, let us bring in leakage.



Equivalent circuit model of a transformer

So for complete modeling, three unknown parameters, L_{lk}, L_m, $(N_1 : N_2)$ turns ratio.

Can we equate this to our mathematical model of the coupled circuit.



$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$v_2 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

Three parameters.

Q. How do we match the circuit model parameters with the mathematical model of the coupled circuit?

Test 1: Set $i_2 = 0$ [open circuit wdg #2].

In the ckt: $v_1 = (L_{lk} + L_m) \frac{di_1}{dt}$

In the mathematical model: $v_1 = L_1 \frac{di_1}{dt}$ $\therefore$ $\boxed{L_1 = L_{lk} + L_m}$

Test 2: Set $i_2 = 0$ [open circuit wdg #2].

In the circuit: $v_2' = L_m \frac{di_1}{dt}$

or, $v_2 = \frac{N_2}{N_1} v_2' = \frac{N_2}{N_1} L_m \frac{di_1}{dt}$

In the mathematical model: $v_2 = M \frac{di_1}{dt}$.

$\therefore$ $\boxed{M = \frac{N_2}{N_1} L_m}$

Test 3: Set $v_2 = 0$ [short circuit wdg #2].

In the circuit: $v_2' = 0 \quad \therefore \quad v_1 = L_{lk} \frac{di_1}{dt}$.

In the mathematical model:

$$v_1 = L_1 \frac{di_1}{dt} + M \frac{di_2}{dt}$$

$$0 = M \frac{di_1}{dt} + L_2 \frac{di_2}{dt}$$

$$\Rightarrow v_1 = \left(L_1 - \frac{M^2}{L_2}\right) \frac{di_1}{dt}$$

$\therefore$ $\boxed{L_{lk} = L_1 - \frac{M^2}{L_2}}$